YEAR 7 MATHEMATICS

Year 7 Revision

Practice Questions

Work in the space under each question. Show your working. Answers are in the separate Answers booklet.

SECTION	CORE QUESTIONS ★	ALL	CORE
A Operations with numbers	1, 5, 6, 9	28 min	15 min
B Computation with integers	1, 5, 6, 8	27 min	14 min
C Indices	1, 4, 7, 9, 10	26 min	13 min
D Fractions, decimals and percentages	4, 5, 9, 10	31 min	14 min
E Algebraic techniques	4, 5, 8, 10	27 min	13 min
F Equations	1, 6, 8, 9, 10	26 min	14 min
G Linear relationships	2, 4, 7, 9	26 min	13 min
H Probability	1, 6, 7, 9	25 min	13 min
I Mixed challenge	3, 6, 8	24 min	11 min
Total		4 h 0 min	2 h 0 min

This pack holds more than one lesson's worth of questions, so choose questions to suit the student. The questions marked ★ make a suggested core set for a 2-hour lesson. The core set leans towards the harder questions. Most questions are taken word for word from ICE-EM Mathematics 7 (3rd edition), and the source is shown beside each one. Questions marked ‘Pack question’ were written for this pack, to cover question types from the Year 7 tasks that the textbook does not have. Times are a guide for a typical Year 7 student.

SECTION A

Operations with numbers

Order of operations; long multiplication; division with remainders; mental strategies; divisibility tests; HCF and LCM.

A OPERATIONS WITH NUMBERS · Y7 ICE-EM 1H, 1L, 2E, 2H, 2I

About 28 minutes. Core: questions 1, 5, 6 and 9.

1 ★ Evaluate: Y7 ICE-EM Ex 1L Q4

a $(4-3)\times 5$

d $(32-16)+(54-12)\div 6$

f $4\times 11\div 2\times(12+8)$

i $(11+7)\div 3+8\times(11+19)$

2 Evaluate: Pack question

a $\dfrac{48-6\times 3}{2+3}$

b $\dfrac{7\times 8-20}{3\times 4}$

3 Carry out each calculation, using the long multiplication method. Y7 ICE-EM Ex 1H Q2

h 5646×73

4 Calculate each division. Express your answer as a mixed numeral in simplest form. Pack question

a $587\div 6$

b $2270\div 12$

5 ★ Show your working to make clear the steps you would take to do these calculations mentally. Y7 ICE-EM Ex 2E Q6

a 125×32 b 135×22

d $1344 \div 16$ e 367×9

6 ★ Which of the numbers below divide the number 584 760? Y7 ICE-EM Ex 2I Q7

2, 3, 4, 5, 6, 8, 9, 10

7 Find the HCF and LCM of: Y7 ICE-EM Ex 2H Q3

a 224 and 336

8 A bell rings every 15 minutes and a whistle is blown every 18 minutes. The bell is rung and the whistle is blown at 8 a.m. How long will it be before the bell is rung and the whistle blown at the same time again? Y7 ICE-EM Ex 2H Q4

9 ★ The five-digit number $\square 725 \triangle$ is divisible by 72. Find the values of $\square$ and $\triangle$. Y7 Ch 2 Challenge Q10

SECTION B

Computation with integers

Adding and subtracting negative integers; multiplying and dividing; order of operations with powers; problems with gains and losses.

B COMPUTATION WITH INTEGERS · Y7 ICE-EM 11C–11F

About 27 minutes. Core: questions 1, 5, 6 and 8.

1 ★ Evaluate: Y7 ICE-EM Ex 11C Q3

c $-39 + 54 - 1$

f $-28 - (-35) - (-2)$

j $-92 + 54 - (-82)$

l $-500 + 2000 - (-50)$

2 The temperature in Canberra had gone down to $-3°$C. The temperature in a heated house was a cosy $22°$C. What was the value of: Y7 Ch 11 Review Q4

a (inside temperature) — (outside temperature)?

b (outside temperature) — (inside temperature)?

3 Evaluate: Y7 ICE-EM Ex 11D Q3

a $3 \times (-2) \times (-6)$

d $-45 \times (-7) \times 20$

f $-34 \times (-3) \times (-2)$

i $-1 \times (-1) \times (-6)$

4 Put multiplication ($\times$) or division ($\div$) signs in each box to make each statement true. Y7 ICE-EM Ex 11E Q6

a $8 \square 4 \square (-2) \square 3 = -12$

b $3 \square 4 \square (-5) \square 6 = -10$

5 ★ Evaluate: Y7 ICE-EM Ex 11F Q9

b $-5 + 49 \div (-7)^2 + 2$

d $3 - 50 \div (3 - 8)^2 - 2$

f $7 - 3^2 \times (1 - 3)^2$

h $16 + 12 \div (-2)^2 - 4$

6 ★ The local charity is planning to run a fair, and is trying to decide how much to charge for entry. The hall where it plans to hold it will cost $\$500$ to rent for the day. It plans to charge $\$5$ per person for entry, and to give each person a show bag that costs $\$2$ to produce. Y7 ICE-EM Ex 11F Q13

a If 120 people come to the fair, how much money will the charity make?

b If the charity decides instead to charge $\$8$ per person, and 120 people attend, how much will it make or lose?

c If it charges $\$5$ per child and $\$8$ per adult, and 60 children and 60 adults attend, how much will it make or lose?

7 Put the three numbers 4, -2 and -7 into the boxes below Y7 Ch 11 Challenge Q4

$$\square + \square - \square = \square$$

so that the answer is:

a -1 b 9 c -13

8 ★ Evaluate $100 + 99 - 98 - 97 + \cdots + 4 + 3 - 2 - 1$. Y7 Ch 11 Challenge Q10

9 The average of five numbers was 2. If the smallest number is deleted, the average is 4. What is the smallest number? Y7 Ch 11 Challenge Q11

SECTION C

Indices

Index notation; multiplying powers of the same number; prime factorisation in index form; squares and square roots; powers of negative numbers and decimals.

C INDICES · Y7 ICE-EM 2D, 2F, 2G, 11F, 14C

About 26 minutes. Core: questions 1, 4, 7, 9 and 10.

1 ★ Rewrite each expression in index notation. Y7 ICE-EM Ex 2D Q7

c $2 \times 2 \times 2 \times 2^2 \times 3 \times 3 \times 3^3$

f $3^2 \times 7^2 \times 3 \times 7^3$

2 Evaluate: Y7 ICE-EM Ex 2D Q9

c 4×5^2

d $(4 \times 5)^2$

f $(2 \times 2)^3$

g $(3 \times 10^3) \times (2 \times 10^2)$

3 Copy and complete each statement. Y7 ICE-EM Ex 2D Q6

a $2^3 \times 2^{\square} = 2^8$

b $3 \times 3^{\square} = 3^{10}$

d $8^{\square} \times 8^5 = 8^6$

f $6 \times 6 \times 6^{\square} = 6^5$

4 ★ Express each number as a product of prime factors. Y7 ICE-EM Ex 2F Q2

c 36

d 96

f 841

5 Find the square root of: Y7 ICE-EM Ex 2G Q2

g 625

i 1444

j 5625

6 Evaluate: Y7 ICE-EM Ex 11F Q4

a $(-8)^2$ b $-(11)^2$ d $-9 \times (-3)^2$ g $(-5)^3$

7 ★ Evaluate: Y7 ICE-EM Ex 11F Q5

c $4 \times (-4)^3$ g $4 \times (-3)^3$ h $7 \times (-1)^{23}$

8 Complete the following calculations. Give your answers in decimal form. Y7 ICE-EM Ex 14C Q5

c $(-0.3)^3 + (-0.4)^3$ f $(-0.2)^3 + (-0.2)^4$

9 ★ Y7 Ch 2 Review Q14

d How many zeros are there in the number $2^3 \times 5^4$?

f How many zeros are there in the number $2^{21} \times 5^{20}$?

g How many zeros are there in the number $2^{23} \times 5^{20}$?

10 ★ Y7 Ch 2 Challenge Q11

a How many factors does $1\,048\,576 = 2^{20}$ have?

b How many factors does $729 = 3^6$ have?

c How many factors does $2^{20} \times 3^6$ have?

SECTION D

Fractions, decimals and percentages

Simplest form; adding, multiplying and dividing fractions; order of operations with fractions; recurring decimals; decimal and negative decimal calculations; percentages of quantities; fraction and percentage problems.

D FRACTIONS, DECIMALS AND PERCENTAGES · Y7 ICE-EM Ch 4, 5, 8, 14, 15

About 31 minutes. Core: questions 4, 5, 9 and 10.

1 Reduce each of these fractions to its simplest form. Y7 ICE-EM Ex 4B Q6

q $\frac{504}{72}$ r $\frac{171}{285}$ t $\frac{42}{91}$ u $\frac{72}{48}$

2 Perform these additions, writing your answers as mixed numerals. Y7 ICE-EM Ex 4E Q4

e $\frac{12}{15} + \frac{3}{5}$ h $\frac{20}{21} + \frac{5}{7}$ k $\frac{2}{3} + \frac{7}{10}$

3 Evaluate: Y7 Ch 5 Review Q2

a $\frac{2}{3} \times \frac{3}{5} \div \frac{1}{4}$ c $6 \div \frac{1}{3} + \frac{1}{2}$ d $1\frac{1}{3} \times 1\frac{1}{4} + 2$

4 ★ Evaluate: Y7 ICE-EM Ex 5E Q2

e $\frac{5}{6} + \left(\frac{2}{3}\right)^2$

j $\frac{7}{8} + \frac{5}{4} \times \frac{4}{15} - \frac{5}{12}$

o $\left(\frac{4}{5} - \frac{2}{3}\right) \times \left(\frac{2}{6} + \frac{5}{8}\right) \div \left(\frac{1}{2} + \frac{2}{3}\right)$

5 ★ Express each fraction as a decimal. Y7 Ch 8 Review Q11

a $\frac{3}{7}$ b $\frac{2}{11}$ d $\frac{6}{13}$ f $\frac{7}{12}$

6 Evaluate the following, giving your answers in decimal form. Y7 Ch 14 Review Q4

f $-9.5 + 3.7 \times (-2.1)$ i $-7 \times (-3.2) + 4$ k $-4.5 \times (-4) \div (-12)$

7 Express each quantity as a percentage of the second. Y7 ICE-EM Ex 15B Q2

e 23 mm, 5 cm i 1250 m, 2 km l $\$34.20$, $\$60$

8 Omar reads $\frac{2}{11}$ of the pages of a book one day, $\frac{3}{5}$ the next day and the remaining 84 pages the following day. How many pages are in the book? Y7 Ch 5 Review Q10

9 ★ Ana went on a shopping spree. She spent $\$24$ of her pocket money on a dress and 20% of the remainder on a shirt. She still had $\frac{2}{3}$ of her money left. How much did she have before she began spending? Y7 Ch 15 Challenge Q4

10 ★ A rectangle has its length decreased by 20%. By what percentage must its width be increased so that the area of the rectangle stays the same? Pack question

SECTION E

Algebraic techniques

Collecting like terms; multiplying terms; order of operations in algebra; expanding brackets and simplifying; writing expressions from words; substitution with fractions and negative numbers.

E ALGEBRAIC TECHNIQUES · Y7 ICE-EM 3D, 3F, 7B, 12A, 16F

About 27 minutes. Core: questions 4, 5, 8 and 10.

1 Simplify each expression by collecting like terms. Y7 Ch 3 Review Q20

a $10a - a + 2b - b$

c $4a^2b + 6a^2b + 11a^2b + a^2b$

d $2c^2d + 2cd + 2dc^2 + 3cd$

2 Simplify: Y7 ICE-EM Ex 3F Q4

a $(2x)^2 \times x^3$

b $3x \times (4x)^2$

f $5z^2y \times 6zy$

3 Simplify: Pack question

a $9p + 24p \div 4$

b $30q \div 5 - 2q \times 2$

4 ★ Simplify each expression by first expanding brackets and then adding or subtracting like terms. Y7 ICE-EM Ex 16F Q2

f $3(2x + 6) + 5(x - 6)$

g $4(2x + 3) + 5(x - 4)$

h $3(4x + 2) + 7(x + 4)$

5 ★ Expand and simplify: Pack question

a $9y^2 + 3y(2y - 7)$

b $4p(p + 3) - 5p$

6 Write each of these expressions using algebraic notation. In each part, use the pronumeral x. Y7 ICE-EM Ex 7B Q1

d A number is multiplied by 5, then 2 is subtracted from the result, and the result of this is then divided by 5.

j 3 is taken away from $\frac{2}{5}$ of a number, and the result of this is then multiplied by 4.

k $\frac{4}{9}$ of a number is subtracted from 23, and the result of this is then multiplied by $\frac{3}{7}$.

7 Buffy has $\$1000$ in a bank account. She takes $\$x$ from the bank account every day. Y7 ICE-EM Ex 12A Q10

a How much money does she have in the account after:

i 1 day? ii 5 days?

b Find the value of her bank account after 5 days if:

i $x = 100$ ii $x = 200$ iii $x = 250$

8 ★ For $m = 6$ and $n = \frac{1}{2}$, evaluate these expressions. Y7 ICE-EM Ex 7B Q10

a $3\left(\frac{m}{n} + 10\right)$ **c** $m + \frac{3m - 34n}{4}$ **d** $\frac{m + 23}{n} - 4m$

9 Evaluate each expression for $w = -2$. Y7 ICE-EM Ex 12A Q9

d $(-2w)^3$ **e** $(-w)^3 + w^2$ **f** $w^3 - 10w^2$

10 ★ The area of the rectangle $ABCD$ is 16 cm^2. The length of BA is x cm. Y7 Ch 7 Challenge Q7



a Find the length of BC, in terms of x.

b Write down an expression for the perimeter of the rectangle, in terms of x.

c If $x = 4$, find the perimeter of the rectangle.

d If $x = \frac{1}{4}$, find the perimeter of the rectangle.

SECTION F

Equations

One-step and two-step equations; fractions in equations; negative solutions; equations with brackets; collecting like terms first; writing and solving equations for problems.

F EQUATIONS · Y7 ICE-EM 16C–16H

About 26 minutes. Core: questions 1, 6, 8, 9 and 10.

1 ★ Solve each equation. Y7 ICE-EM Ex 16C Q3

d $4x + 18 = 30$

j $\frac{x}{12} - 2 = 15$

l $\frac{z}{7} + 6 = 18$

2 Solve: Y7 Ch 16 Review Q2

e $\frac{5x}{4} = 12$

k $\frac{x}{10} = -2$

p $5(3x - 5) = 17$

3 Solve: Pack question

a $\frac{x + 7}{4} = 6$

b $\frac{2y - 1}{5} = 3$

4 Solve: Y7 Ch 16 Review Q6

g $\frac{-3x}{2} = -8$

h $4x + 1 = -8$

i $-2x + 10 = -12$

5 Expand the brackets and solve each equation. Y7 ICE-EM Ex 16E Q3

d $6(x - 7) = 12$

f $7(4n - 10) = 11$

6 ★ Solve each equation by first collecting like terms. Y7 ICE-EM Ex 16G Q3

b $60 - 2x + 8x = 4x + 100$ d $60 = 7x + 45 - 4x$

7 A rectangle has width 4 cm and length $(3x + 2)$ cm. Pack question

a Write an expression for the area of the rectangle.

b The area of the rectangle is 50 cm^2. Find x.

8 ★ $\$400$ is to be shared among three friends. David gets $\$8$ more than Jennifer who gets $\$16$ more than Brett. How much does each of the friends receive? Y7 Ch 16 Review Q14

9 ★ Renae and James have $\$60$ between them. Renae gives James $\$12$. She now has twice as much as James. How much did she have previously? Y7 Ch 16 Challenge Q6

10 ★ Three boxes are weighed two at a time. Boxes A and B together weigh 27 kg, boxes B and C together weigh 33 kg, and boxes A and C together weigh 30 kg. How much does each box weigh? Pack question

SECTION G

Linear relationships

Number patterns; reading and plotting points; tables of values; finding rules from tables, patterns and graphs; using rules; lines and the axes.

G LINEAR RELATIONSHIPS · Y7 ICE-EM 12B–12D

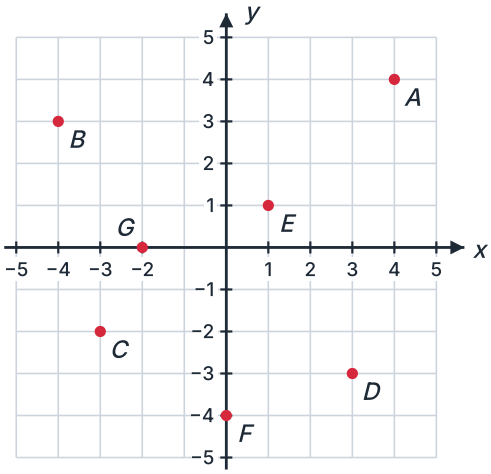
About 26 minutes. Core: questions 2, 4, 7 and 9.

1 Write down the next five numbers, continuing the obvious pattern in each case. Y7 Ch 8 Challenge Q1

c 12.5, 12.4, 12.3, ___, ___, ___, ___, ___

d 12.4, 11.3, 10.2, ___, ___, ___, ___, ___

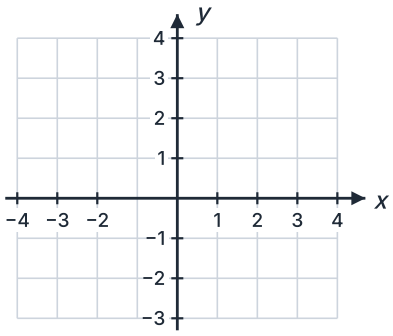
2 ★ Write down the coordinates of each of the points A to G. Y7 Ch 12 Review Q3



3 For each given rule, complete the table, list the coordinates, and plot the corresponding set of points on a number plane. Y7 ICE-EM Ex 12C Q2

d $y = -x + \frac{1}{2}$

x	-3	-2	-1	0	1	2	3
y							



4 ★ A pile of matchsticks is used to make the following pattern of shapes. The first diagram uses three matches to form one triangle. The second diagram uses five matches to form two triangles. Y7 ICE-EM Ex 12D Q2



Diagram 1 Diagram 2 Diagram 3 Diagram 4

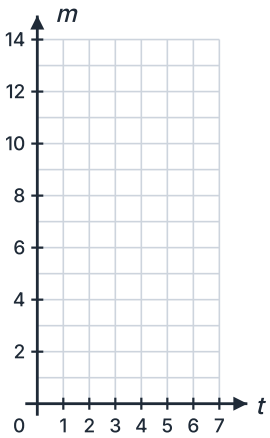
a Count the number of matches used to make each diagram, and complete the table below.

Number of triangles (t)	1	2	3	4	5	6
Number of matches (m)	3	5	7			

b How many matches do we add each time to create an extra triangle?

c Plot the points (t, m) for values of t from 1 to 6, using your table of values.

d Write a rule that tells us the number, m, of matches we need to make any number of triangles.



5 Fill in the boxes to give a rule for each of the following tables. Y7 Ch 12 Review Q10

c

m	1	2	3	4	5
n	0	4	8	12	16

$n = \square \times m - \square$

d

p	-2	-1	0	1	2
q	-5	-3	-1	1	3

$q = \square \times p - \square$

f

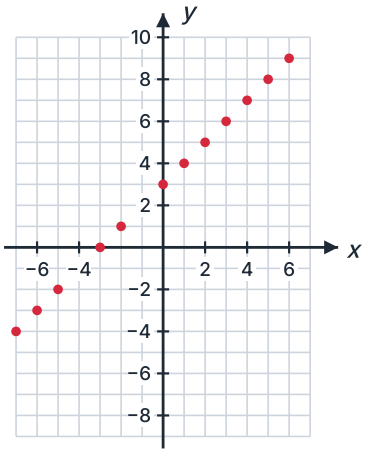
x	-2	-1	0	1	2
y	-1	-4	-7	-10	-13

$y = \square \times x - \square$

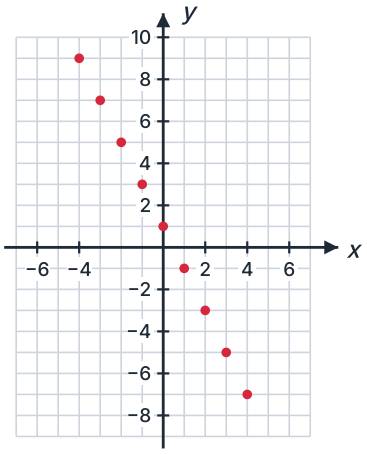
6 Match each diagram with the correct rule from the list below. Y7 ICE-EM Ex 12D Q8

$y = 3x$ $y = -2x$ $y = x + 3$ $y = -2x - 1$ $y = -2x + 1$ $y = x - 4$

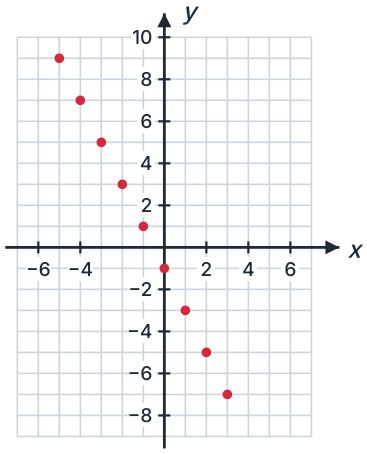
a



c



f



7 ★ Sarah is given $\$1000$ for her 18th birthday. She decides to use it to sponsor a child in Africa at a cost of $\$20$ each month. Y7 ICE-EM Ex 12D Q6

a Complete the table below to show how much money Sarah has left at the end of each month.

Month (m)	0	1	2	3	4	5	6
Dollars (d)	1000	980					

b Write a rule to show how many dollars, d, Sarah has left after m months.

c How much money will Sarah have after 10 months?

d For how many months can Sarah sponsor the child?

8 The number plane axes have markers at 1 cm intervals; that is, the point $(1, 0)$ is 1 cm from the origin. AB is an interval on the number plane. A has coordinates $(5, 0)$ and B has coordinates $(10, 0)$. Describe the points C such that triangle ABC has area 20 cm^2. Y7 Ch 12 Challenge Q4

9 ★ The line with rule $y = 2x + c$ passes through the point $(3, 2)$. Pack question

a Find the value of c.

b Find the coordinates of the points where the line crosses the x-axis and the y-axis.

c Find the area of the triangle formed by the line and the two axes.

SECTION H

Probability

Probability from 0 to 1; sample spaces; equally likely outcomes; ‘and’, ‘or’ and ‘not’; two-way tables and Venn diagrams; working backwards from a probability.

H PROBABILITY · Y7 ICE-EM 17A–17B

About 25 minutes. Core: questions 1, 6, 7 and 9.

1 ★ Complete each of the following probability statements. Y7 ICE-EM Ex 17A Q1

c If there are three red discs and three blue discs in a bag and you take one out without looking in the bag, you are equally likely to get a red disc or a blue disc.
If there are three red discs and three blue discs in a bag and you take one out without looking in the bag, the probability of getting a red disc is _____ and the probability of getting a blue disc is _____

d If it is Thursday today, tomorrow will be Friday.
If it is Thursday today, the probability that it will be Friday tomorrow is _____

e If today is the 31st of January, there is no chance that tomorrow will be the 1st of May. If today is the 31st of January, the probability that tomorrow is the 1st of May is _____

2 A bag contains 10 marbles numbered 1 to 10. A marble is taken out. Y7 ICE-EM Ex 17B Q1

a Give the sample space for this experiment.

b Write down the outcomes favourable to the event ‘a marble with an odd number is taken out’.

3 A bag of sweets contains 10 red ones, nine green ones, six yellow ones and five blue ones. They are all the same size and shape. You reach in and take one out at random. Find the probability that the sweet you pick is: Y7 ICE-EM Ex 17B Q12

b green or yellow **c** not red **e** not yellow or blue

4 Twenty cards are numbered 1 to 20. One card is chosen at random. Find the probability that the number on the card is: Pack question

a odd and a multiple of 3 **b** odd or a multiple of 3 (or both) **c** at least 15

5 A bowl contains green and red normal jelly beans and green and red double-flavoured jelly beans. The numbers of the different colours and types are given in the table below. Y7 ICE-EM Ex 17B Q14

	Green	Red
Normal jelly beans	250	400
Double-flavoured jelly beans	175	75

A jelly bean is randomly taken out of the bowl. Find the probability that:

a it is a double-flavoured jelly bean

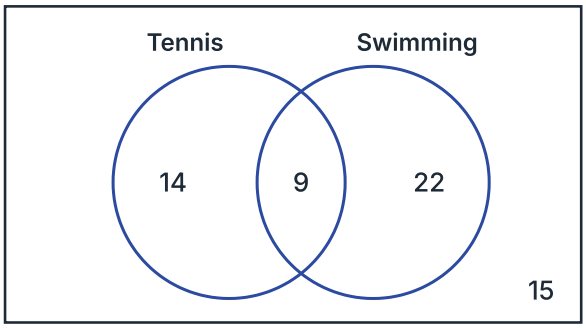
b it is a green jelly bean

c it is a green normal jelly bean

d it is a red double-flavoured jelly bean

e it is a red jelly bean

6 ★ Sixty students were asked whether they play tennis and whether they swim. The Venn diagram shows the results. Pack question



a Complete the two-way table.

	Swims	Does not swim	Total
Plays tennis			
Does not play tennis			
Total			

b A student is chosen at random. Find the probability that the student:

i swims but does not play tennis

ii neither plays tennis nor swims

c Some more students who play tennis but do not swim are added to the survey. Now the probability that a student chosen at random plays tennis is $\frac{1}{2}$. How many students were added?

7 ★ A bag initially contains 15 red balls and 8 blue balls. Some of the red balls are removed. A ball is then drawn out at random. The probability that it is a red ball is $\frac{1}{3}$. How many red balls were removed? Y7 Ch 17 Challenge Q5

8 In a raffle, 50 tickets are sold. The tickets are numbered from 1 to 50. They are placed in a hat and one is drawn out at random to win a prize. Find the probability that the number on the winning ticket: Y7 Ch 17 Challenge Q1

f contains the digit 7

h contains the digit 2 at least once

i contains the digit 2 only once

j does not contain the digit 2

9 ★ A spinner has four sections coloured red, blue, green and yellow. It is known that $P(\text{red}) = \frac{1}{3}$, $P(\text{blue}) = P(\text{green})$ and $P(\text{yellow}) = 2 \times P(\text{blue})$. Find the size of the angle at the centre of the yellow section. Pack question

SECTION I

Mixed challenge

Review and problem-solving questions from the two Year 7 review chapters, on the topics of this pack.

I MIXED CHALLENGE · Y7 ICE-EM 10A, 10B, 22A

About 24 minutes. Core: questions 3, 6 and 8.

1 Evaluate: Y7 ICE-EM Ex 10A Ch 1 Q16

a $5 \times 4 + 6 \div 2 - 1$ b $5 \times (4 + 6) \div 2 - 1$ c $5 \times 4 + (8 \div 2 - 1)$

2 Arrange the following from smallest to largest. Y7 ICE-EM Ex 22A Ch 15 Q4

a $0.39,\ \dfrac{12}{32},\ 4\frac{1}{2}\%$ b $64\%,\ 0.6,\ \dfrac{2}{3}$

3 ★ Evaluate: Y7 ICE-EM Ex 10A Ch 5 Q6

a $\dfrac{\frac{2}{3} + \frac{3}{4}}{\frac{5}{6} - \frac{2}{3}}$ b $\dfrac{\frac{5}{12} - \frac{1}{4}}{\frac{2}{3} + \frac{1}{6}}$ c $\dfrac{\frac{3}{4} \times \frac{8}{9}}{\frac{2}{3} - \frac{1}{6}}$

4 If I spend one-third of my money, then one-quarter of what is left, and finally one-fifth of the remainder, what fraction of my original money do I have left? Y7 ICE-EM Ex 10A Ch 5 Q9

5 Four bells start tolling together and ring every 1, $1\frac{1}{4}$, $1\frac{1}{5}$ and $1\frac{1}{6}$ seconds, respectively. After what period of time will they first ring all together again? Y7 ICE-EM Ex 10A Ch 5 Q10

6 ★ If you add 13 to a number, you get the same result as when you subtract half the number from 4. What is the number? Y7 ICE-EM Ex 22A Ch 16 Q9

7 A box contains 24 balls. There are eight red balls numbered 1 to 8, six blue balls numbered 9 to 14, and ten green balls numbered 15 to 24. One ball is taken out of the box. Find the probability of getting: Y7 ICE-EM Ex 22A Ch 17 Q2

d a green ball with an odd number

e a green or a red ball

g a number divisible by 5

h a prime number

8 ★ Two whole numbers are added together. If their sum is odd, which statements below are always true? Which are always false? Which are sometimes true and sometimes false? Y7 ICE-EM Ex 10B Evens and odds

1 Their quotient is not a whole number.

2 Their product is even.

3 Their difference is even.

4 Their product is more than their sum.

5 If 1 is added to one of the numbers and the product is found, it will be even.